

LANCANG, China

WMO: 569540

Lat: 22.567N

Lon: 99.933E

Elev: 1054

StdP: 89.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.6	(c) 6.6	(d) 1.8	(e) 4.9	(f) 23.0	(g) 3.4	(h) 5.5	(i) 19.6	(j) 6.1	(k) 23.2	(l) 5.3	(m) 23.0	(n) 0.2	(o) 160	(p) 0.233

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.1	(c) 32.5	(d) 19.3	(e) 31.5	(f) 19.4	(g) 30.5	(h) 19.7	(i) 23.2	(j) 27.8	(k) 22.9	(l) 27.3	(m) 22.6	(n) 26.7	(o) 3.1	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	19.1	24.5	21.8	18.7	24.2	21.6	18.5	24.0	75.4	28.0	74.0	27.5	72.8	27.0	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.3	(b) 4.4	(c) 3.6	(e) 3.3	(f) 34.3	(g) 2.0	(h) 1.1	(i) 1.9	(j) 35.0	(k) 0.7	(l) 35.7	(m) -0.4	(n) 36.3	(o) -1.8	(p) 37.0		
			DB													
			WB													
				3.0	23.9	2.0	0.4	1.6	24.2	0.4	24.5	-0.7	24.7	-2.1	25.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.3	15.7	17.1	19.9	22.7	24.2	24.7	24.2	24.4	24.0	22.4	19.2	16.4		
	DBStd	3.63	1.44	1.54	1.58	1.62	1.79	1.27	1.36	1.27	1.52	1.61	1.88	2.00		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	206	82	40	5	1	1	0	0	0	0	1	12	66		
	CDD10.0	4109	177	198	306	381	441	441	440	447	419	384	277	198		
	CDD18.3	1273	1	5	52	132	183	191	182	188	169	127	39	5		
	CDH23.3	8214	61	272	763	1299	1409	1047	807	908	839	568	195	45		
(14) Wind	WSAvg	1.1	1.1	1.3	1.4	1.4	1.3	1.1	1.0	1.0	1.1	1.0	0.9	0.9		
	PrecAvg	1639	18	15	14	43	161	257	343	315	200	153	84	29		
	PrecMax	2099	86	40	83	118	335	464	504	523	448	381	257	130		
	PrecMin	1355	0	0	0	4	35	131	174	134	81	17	5	2		
	PrecStd	202	22	13	17	28	84	85	80	108	79	80	65	31		
	DB	26.1	29.1	31.5	33.6	34.3	32.6	31.3	31.6	31.2	30.1	28.1	25.9			
	MCWB	14.3	15.0	16.0	18.0	19.3	21.5	22.8	22.3	21.9	20.9	19.0	17.0			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	24.9	27.7	30.4	32.4	32.7	30.8	29.7	30.1	30.0	28.9	26.7	24.6			
	2% MCWB	14.0	14.7	15.8	17.8	19.7	21.8	22.6	22.4	21.8	20.3	18.1	16.0			
	5% DB	23.9	26.6	29.3	31.3	31.4	29.4	28.5	28.9	29.0	27.9	25.6	23.6			
	5% MCWB	14.0	14.3	15.6	17.6	19.8	22.0	22.4	22.4	21.6	20.1	17.7	15.5			
	10% DB	22.6	25.2	28.0	30.1	29.9	28.1	27.2	27.6	27.7	26.7	24.4	22.2			
	10% MCWB	13.7	13.9	15.4	17.6	20.0	22.0	22.2	22.2	21.5	20.1	17.3	15.1			
	WB	16.9	16.9	18.6	20.5	22.5	23.5	23.7	23.7	23.3	22.4	20.5	18.9			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	21.6	25.3	25.3	27.4	27.1	28.3	28.6	28.5	28.1	27.0	24.7	22.7			
	2% WB	15.9	15.8	17.6	19.7	22.0	23.0	23.2	23.2	22.8	21.7	19.7	17.8			
	2% MCDB	21.2	24.6	24.6	26.8	27.1	27.4	27.6	27.8	27.2	26.3	23.7	21.8			
	5% WB	15.0	15.1	17.0	19.1	21.6	22.7	22.7	22.7	22.4	21.2	19.2	16.7			
	5% MCDB	21.0	23.7	25.0	26.6	26.5	26.7	26.7	27.0	26.6	25.6	23.0	20.9			
	10% WB	14.3	14.5	16.3	18.6	21.2	22.3	22.4	22.4	22.1	20.8	18.5	15.8			
	10% MCDB	20.4	22.9	24.7	26.4	25.9	26.1	26.0	26.3	26.0	24.8	22.2	20.2			
(35) Mean Daily Temperature Range	MDBR	15.4	17.3	16.8	14.8	10.6	7.1	6.0	6.8	7.9	9.1	11.1	12.6			
	5% DB MCDBR	17.5	19.5	19.0	16.9	14.1	9.7	8.4	9.0	9.9	11.1	12.5	14.1			
	5% WB MCWBR	7.4	7.1	5.7	4.4	3.3	2.4	2.3	2.5	2.8	3.5	4.7	6.1			
	5% DB MCDBR	14.2	16.7	15.0	13.4	9.2	7.4	6.9	7.6	8.1	8.7	9.4	11.3			
	5% WB MCWBR	6.4	6.5	5.1	4.0	2.6	2.2	2.2	2.4	2.5	3.0	3.7	5.1			
(40) Clear-Sky Solar Irradiance	taub	0.357	0.417	0.673	0.652	0.474	0.426	0.422	0.455	0.454	0.448	0.382	0.366			
	taud	2.351	2.179	1.546	1.650	2.198	2.386	2.385	2.264	2.242	2.220	2.346	2.355			
	Ebn at Noon	906	868	672	699	831	864	868	843	840	827	872	882			
	Edh at Noon	113	143	282	258	148	122	122	138	138	136	113	109			
(44) All-Sky Solar Radiation		RadAvg	4.30	5.29	5.43	5.65	5.12	4.41	3.98	4.28	4.36	4.08	4.12	3.77		
(45)		RadStd	0.28	0.34	0.32	0.41	0.50	0.37	0.30	0.31	0.28	0.37	0.31	0.30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.34	+0.51	+0.64	N/A	N/A	N/A	N/A	N/A	+140	+127
(47) Regional (6 neighbors)		+0.32	N/A	N/A	+0.50	N/A	N/A	N/A	-44	+144	+105

Nomenclature: See separate page